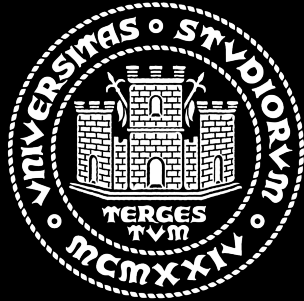




Stochastic Schrödinger Equation - Numerical Methods

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Chapter 1

Time-Domain Propagation

The time-dependent Schrödinger equation (TDSE) including the perturbation due to an external electric field $\mathbf{F}(t)$ reads

$$\begin{aligned} i\frac{d}{dt}|\Psi(t)\rangle &= \hat{H}(t)|\Psi(t)\rangle \\ &= \left(\hat{H}_0 + \hat{H}_I(t)\right)|\Psi(t)\rangle \\ &= \left(\hat{H}_0 - \mathbf{F}(t) \cdot \hat{\boldsymbol{\mu}}\right)|\Psi(t)\rangle, \end{aligned} \quad (1.1)$$

where $\hat{H}_0$ is the field-free Hamiltonian, $\hat{H}_I(t)$ is the perturbation Hamiltonian, $\hat{\boldsymbol{\mu}}$ is the dipole moment operator, and $|\Psi(t)\rangle$ is the time-dependent electronic wavefunction at time t .

Expanding the wavefunction in the basis of eigenstates $|\lambda\rangle$ of the field-free Hamiltonian (as obtained, e.g., from linear-response TDDFT), we write

$$|\Psi(t)\rangle = \sum_{\lambda}^N C_{\lambda}(t) |\lambda\rangle, \quad (1.2)$$

where N is the number of states retained. Substituting (1.2) into the TDSE yields

$$\sum_{\lambda}^N i\frac{dC_{\lambda}(t)}{dt} |\lambda\rangle = \sum_{\lambda}^N (\hat{H}_0 - \mathbf{F}(t) \cdot \hat{\boldsymbol{\mu}}) C_{\lambda}(t) |\lambda\rangle. \quad (1.3)$$

Projecting onto $\langle\lambda'|$ yields

$$i\frac{dC_{\lambda'}(t)}{dt} = \sum_{\lambda}^N \langle\lambda'|\hat{H}_0 - \mathbf{F}(t) \cdot \hat{\boldsymbol{\mu}}|\lambda\rangle C_{\lambda}(t) \quad (1.4)$$

$$= E_{\lambda'} C_{\lambda'}(t) - \sum_{\lambda}^N \langle\lambda'|\mathbf{F}(t) \cdot \hat{\boldsymbol{\mu}}|\lambda\rangle C_{\lambda}(t), \quad (1.5)$$

where $E_{\lambda'}$ is the energy of $|\lambda'\rangle$. In the one-hole-one-particle (1h-1p) space each $|\lambda\rangle$ expands as

$$|\lambda\rangle = \sum_i^{N_{occ}} \sum_a^{N_{virt}} d_{i,\lambda}^a |\phi_i^a\rangle, \quad (1.6)$$

where N_{occ} and N_{virt} denote the numbers of occupied and virtual orbitals, and $d_{i,\lambda}^a$ expands $|\lambda\rangle$ in the singly excited determinants $|\phi_i^a\rangle$ (promotion from occupied i to virtual a). The TDSE in the 1h-1p space then becomes

$$\begin{aligned} i\frac{dC_{\lambda'}(t)}{dt} &= E_{\lambda'}C_{\lambda'}(t) - \sum_{\lambda}^N \sum_{i,j}^{N_{\text{occ}}} \sum_{a,b}^{N_{\text{virt}}} d_{i,\lambda'}^a d_{j,\lambda}^b \langle \phi_i^a | \mathbf{F}(t) \cdot \hat{\boldsymbol{\mu}} | \phi_j^b \rangle C_{\lambda}(t) \\ &= E_{\lambda'}C_{\lambda'}(t) - \sum_{q=x,y,z}^N \sum_{\lambda}^{N_{\text{occ}}} \sum_{i,j}^{N_{\text{virt}}} d_{i,\lambda'}^a d_{j,\lambda}^b F_q(t) \langle \phi_i^a | \hat{\mu}_q | \phi_j^b \rangle C_{\lambda}(t), \end{aligned} \quad (1.7)$$

where $F_q(t)$ is the q -th component of $\mathbf{F}(t)$ and $\hat{\mu}_q$ is that of $\hat{\boldsymbol{\mu}}$. The Slater-Condon rules give

$$\langle \phi_i^a | \hat{\boldsymbol{\mu}} | \phi_j^b \rangle = \begin{cases} \mathbf{D}_0 - \boldsymbol{\mu}_{ii} + \boldsymbol{\mu}_{aa} & \text{if } i = j, a = b, \\ \boldsymbol{\mu}_{ab} & \text{if } i = j, a \neq b, \\ -\boldsymbol{\mu}_{ij} & \text{if } i \neq j, a = b, \\ 0 & \text{otherwise,} \end{cases} \quad (1.8)$$

where $\boldsymbol{\mu}_{lm}$ is the dipole matrix element in the molecular-orbital basis (as supplied by standard quantum-chemistry codes), and $\mathbf{D}_0 = \sum_l \boldsymbol{\mu}_{ll}$. In matrix form, the TDSE in the 1h-1p space reads

$$i\frac{d\mathbf{C}(t)}{dt} = \mathbf{H}(t)\mathbf{C}(t), \quad (1.9)$$

where $\mathbf{C}(t)$ collects the expansion coefficients and $\mathbf{H}(t)$ is the Hamiltonian matrix in the 1h-1p basis, with elements

$$(\mathbf{H}(t))_{\lambda',\lambda} = E_{\lambda}\delta_{\lambda',\lambda} - \sum_{q=x,y,z}^N \sum_{i,j}^{N_{\text{occ}}} \sum_{a,b}^{N_{\text{virt}}} d_{i,\lambda'}^a d_{j,\lambda}^b F_q(t) \langle \phi_i^a | \hat{\mu}_q | \phi_j^b \rangle. \quad (1.10)$$

Chapter 2

Stochastic Schrödinger Equation

In the Markovian limit, the stochastic Schrödinger equation (SSE) reads

$$i \frac{d}{dt} |\Psi_s(t)\rangle = \hat{H}(t) |\Psi_s(t)\rangle + \sum_m^M W_m(t) \hat{S}_m |\Psi_s(t)\rangle - \frac{i}{2} \sum_m^M \hat{S}_m^\dagger \hat{S}_m |\Psi_s(t)\rangle, \quad (2.1)$$

where $\hat{S}_m$ describes coupling to the environment through M channels indexed by m . Dissipation enters via the anti-Hermitian Lindblad contribution $-\frac{i}{2} \sum_m^M \hat{S}_m^\dagger \hat{S}_m$, while the stochastic driving $\sum_m^M W_m(t) \hat{S}_m$ is modeled with Wiener processes $W_m(t)$.

Expanding $|\Psi_s(t)\rangle$ in the same eigenbasis $\{|\lambda\rangle\}$ as in (1.2),

$$|\Psi_s(t)\rangle = \sum_{\lambda}^N C_{\lambda}(t) |\lambda\rangle. \quad (2.2)$$

Substituting (2.2) into the SSE yields

$$\sum_{\lambda}^N i \frac{dC_{\lambda}(t)}{dt} |\lambda\rangle = \sum_{\lambda}^N C_{\lambda}(t) \hat{H}(t) |\lambda\rangle + \sum_{\lambda}^N \sum_m^M W_m(t) C_{\lambda}(t) \hat{S}_m |\lambda\rangle - \frac{i}{2} \sum_{\lambda}^N \sum_m^M C_{\lambda}(t) \hat{S}_m^\dagger \hat{S}_m |\lambda\rangle. \quad (2.3)$$

Hence, projecting onto $\langle\lambda'|\$ gives

$$i \frac{dC_{\lambda'}(t)}{dt} = \sum_{\lambda}^N (\mathbf{H}(t))_{\lambda',\lambda} C_{\lambda}(t) + \sum_{\lambda}^N \sum_m^M W_m(t) (\mathbf{S}_m)_{\lambda',\lambda} C_{\lambda}(t) - \frac{i}{2} \sum_{\lambda}^N \sum_m^M (\mathbf{\Gamma}_m)_{\lambda',\lambda} C_{\lambda}(t), \quad (2.4)$$

with $(\mathbf{S}_m)_{\lambda',\lambda} = \langle\lambda'|\hat{S}_m|\lambda\rangle$, $(\mathbf{\Gamma}_m)_{\lambda',\lambda} = \langle\lambda'|\hat{S}_m^\dagger \hat{S}_m|\lambda\rangle$, and $(\mathbf{H}(t))_{\lambda',\lambda}$ as in (1.10).

In compact matrix notation,

$$\frac{d\mathbf{C}(t)}{dt} = \left(-i\mathbf{H}(t) - \frac{1}{2} \sum_m^M \mathbf{\Gamma}_m \right) \mathbf{C}(t) - i \sum_m^M W_m(t) \mathbf{S}_m \mathbf{C}(t). \quad (2.5)$$

If excited state relaxation from a state $|\lambda_1\rangle$ to a state $|\lambda_2\rangle$ is to be considered in this framework, the operator $\hat{S}_m$ should be replaced by $\hat{S}_{\lambda_1, \lambda_2}$, defined by:

$$\hat{S}_{\lambda_1, \lambda_2} = \sqrt{\Gamma_{\lambda_2, \lambda_1}} |\lambda_2\rangle \langle \lambda_1|. \quad (2.6)$$

where $\Gamma_{\lambda_2, \lambda_1}$ is the transition rate from $|\lambda_1\rangle$ to $|\lambda_2\rangle$. This means that the matrix $\mathbf{S}_m$ should be replaced by $\mathbf{S}_{\lambda_1, \lambda_2}$, defined by:

$$\begin{aligned} (\mathbf{S}_{\lambda_1, \lambda_2})_{\lambda', \lambda} &= \langle \lambda' | \hat{S}_{\lambda_1, \lambda_2} | \lambda \rangle \\ &= \sqrt{\Gamma_{\lambda_2, \lambda_1}} \langle \lambda' | \lambda_2 \rangle \langle \lambda_1 | \lambda \rangle \\ &= \sqrt{\Gamma_{\lambda_2, \lambda_1}} \delta_{\lambda', \lambda_2} \delta_{\lambda_1, \lambda}. \end{aligned} \quad (2.7)$$

In matrix form, $\mathbf{S}_{\lambda_1, \lambda_2}$ is zero everywhere except $\sqrt{\Gamma_{\lambda_2, \lambda_1}}$ in row λ_2 and column λ_1 :

$$\mathbf{S}_{\lambda_1, \lambda_2} = \begin{matrix} & & \lambda_1 & & \\ & & 0 & \dots & 0 \\ \lambda_2 & \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \sqrt{\Gamma_{\lambda_2, \lambda_1}} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix} & & \end{matrix} \quad (2.8)$$

For the matrix $\mathbf{\Gamma}_m$, it should be replaced by $\mathbf{\Gamma}_{\lambda_1, \lambda_2}$, defined by:

$$\begin{aligned} (\mathbf{\Gamma}_{\lambda_1, \lambda_2})_{\lambda', \lambda} &= \langle \lambda' | \hat{S}_{\lambda_1, \lambda_2}^\dagger \hat{S}_{\lambda_1, \lambda_2} | \lambda \rangle \\ &= \Gamma_{\lambda_2, \lambda_1} \langle \lambda' | \lambda_1 \rangle \langle \lambda_2 | \lambda_2 \rangle \langle \lambda_1 | \lambda \rangle \\ &= \Gamma_{\lambda_2, \lambda_1} \langle \lambda' | \lambda_1 \rangle \langle \lambda_1 | \lambda \rangle \\ &= \Gamma_{\lambda_2, \lambda_1} \delta_{\lambda', \lambda_1} \delta_{\lambda_1, \lambda}. \end{aligned} \quad (2.9)$$

In matrix form, $\mathbf{\Gamma}_{\lambda_1, \lambda_2}$ is zero everywhere except $\Gamma_{\lambda_2, \lambda_1}$ in row λ_1 and column λ_1 :

$$\mathbf{\Gamma}_{\lambda_1, \lambda_2} = \begin{matrix} & & \lambda_1 & & \\ & & 0 & \dots & 0 \\ \lambda_1 & \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \Gamma_{\lambda_2, \lambda_1} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix} & & \end{matrix} \quad (2.10)$$

The SSE then becomes:

$$\frac{d\mathbf{C}(t)}{dt} = \left(-i\mathbf{H}(t) - \frac{1}{2} \sum_{\lambda_1, \lambda_2} \mathbf{\Gamma}_{\lambda_2, \lambda_1} \right) \mathbf{C}(t) - i \sum_{\lambda_1, \lambda_2} W_{\lambda_1, \lambda_2}(t) \mathbf{S}_{\lambda_1, \lambda_2} \mathbf{C}(t), \quad (2.11)$$

where $W_{\lambda_2, \lambda_1}(t)$ is the Wiener process for the transition from $|\lambda_1\rangle$ to $|\lambda_2\rangle$.

In differential form, the SSE reads:

$$d\mathbf{C}(t) = \left(-i\mathbf{H}(t) - \frac{1}{2} \sum_{\lambda_1, \lambda_2} \mathbf{\Gamma}_{\lambda_2, \lambda_1} \right) \mathbf{C}(t) dt - i \sum_{\lambda_1, \lambda_2} \mathbf{S}_{\lambda_1, \lambda_2} \mathbf{C}(t) dW_{\lambda_1, \lambda_2}(t). \quad (2.12)$$

In Itô notation, the SSE reads:

$$d\mathbf{C}_t = \left(-i\mathbf{H}_t - \frac{1}{2} \sum_{\lambda_1, \lambda_2} \mathbf{\Gamma}_{\lambda_2, \lambda_1} \right) \mathbf{C}_t dt - i \sum_{\lambda_1, \lambda_2} \mathbf{S}_{\lambda_1, \lambda_2} \mathbf{C}_t dW_{\lambda_1, \lambda_2, t}, \quad (2.13)$$

where the subscript t indicates that the quantities are evaluated at time t . The Euler-Maruyama approximation to the true solution $\mathbf{C}$ of the Itô differential equation in Eq. (2.13) is the Markov chain $\bar{\mathbf{C}}$ given by:

$$\bar{\mathbf{C}}_{n+1} = \bar{\mathbf{C}}_n + \left(-i\mathbf{H}_n - \frac{1}{2} \sum_{\lambda_1, \lambda_2} \mathbf{\Gamma}_{\lambda_2, \lambda_1} \right) \bar{\mathbf{C}}_n \Delta t - i \sum_{\lambda_1, \lambda_2} \mathbf{S}_{\lambda_1, \lambda_2} \bar{\mathbf{C}}_n \Delta W_{\lambda_1, \lambda_2, n}, \quad (2.14)$$

where Δt is the time step, n is the time step index, and $\Delta W_{\lambda_1, \lambda_2, n}$ is the increment of the Wiener process. This increment is given by:

$$\Delta W_{\lambda_1, \lambda_2, n} = \sqrt{\Delta t} \xi_{\lambda_1, \lambda_2, n}, \quad (2.15)$$

where $\xi_{\lambda_1, \lambda_2, n} \sim \mathcal{N}(0, 1)$ are independent standard normal random variables.

Then,

$$\bar{\mathbf{C}}_{n+1} = \bar{\mathbf{C}}_n + \left(-i\mathbf{H}_n - \frac{1}{2} \sum_{\lambda_1, \lambda_2} \mathbf{\Gamma}_{\lambda_2, \lambda_1} \right) \bar{\mathbf{C}}_n \Delta t - i\sqrt{\Delta t} \sum_{\lambda_1, \lambda_2} \mathbf{S}_{\lambda_1, \lambda_2} \xi_{\lambda_1, \lambda_2, n} \bar{\mathbf{C}}_n. \quad (2.16)$$

In matrix form, the term $\sum_{\lambda_1, \lambda_2} \mathbf{S}_{\lambda_1, \lambda_2} \xi_{\lambda_1, \lambda_2, n}$ is given by:

$$\begin{aligned} \mathbf{R}_n &= \sum_{\lambda_1, \lambda_2} \mathbf{S}_{\lambda_1, \lambda_2} \xi_{\lambda_1, \lambda_2, n} \\ &= \lambda_2 \begin{pmatrix} \sqrt{\Gamma_{0,0}} \xi_{0,0,n} & \cdots & \sqrt{\Gamma_{0,\lambda_1}} \xi_{0,\lambda_1,n} & \cdots & \sqrt{\Gamma_{0,N}} \xi_{0,N,n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \sqrt{\Gamma_{\lambda_2,0}} \xi_{\lambda_1,0,n} & \cdots & \sqrt{\Gamma_{\lambda_2,\lambda_1}} \xi_{\lambda_1,\lambda_2,n} & \cdots & \sqrt{\Gamma_{\lambda_2,N}} \xi_{\lambda_1,N,n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \sqrt{\Gamma_{N,0}} \xi_{N,0,n} & \cdots & \sqrt{\Gamma_{N,\lambda_1}} \xi_{N,\lambda_1,n} & \cdots & \sqrt{\Gamma_{N,N}} \xi_{N,N,n} \end{pmatrix}. \end{aligned} \quad (2.17)$$

The term $\sum_{\lambda_1, \lambda_2} \mathbf{\Gamma}_{\lambda_2, \lambda_1}$ is given by:

$$\begin{aligned} \mathbf{\Gamma} &= \sum_{\lambda_1, \lambda_2} \mathbf{\Gamma}_{\lambda_2, \lambda_1} \\ &= \begin{pmatrix} \sum_{\lambda_2} \Gamma_{\lambda_2,0} & 0 & \cdots & \cdots & 0 \\ 0 & \sum_{\lambda_2} \Gamma_{\lambda_2,1} & 0 & \cdots & 0 \\ 0 & \cdots & \ddots & \cdots & 0 \\ 0 & \cdots & \cdots & \sum_{\lambda_2} \Gamma_{\lambda_2,N-1} & 0 \\ 0 & \cdots & \cdots & \cdots & \sum_{\lambda_2} \Gamma_{\lambda_2,N} \end{pmatrix} \end{aligned} \quad (2.18)$$

Consequently, the Euler-Maruyama approximation to the SSE is given by:

$$\bar{\mathbf{C}}_{n+1} = \bar{\mathbf{C}}_n + \Delta t \left(-i\mathbf{H}_n - \frac{1}{2} \mathbf{\Gamma} \right) \bar{\mathbf{C}}_n - i\sqrt{\Delta t} \mathbf{R}_n \bar{\mathbf{C}}_n. \quad (2.19)$$

Chapter 3

Weighted Normalization Scheme

When the typical wavefunction normalization scheme (i.e., the squared moduli of the coefficients are rescaled so that their sum equals 1) is applied at each time step of the Euler-Maruyama approximation, noise can enter coefficients that would otherwise evolve deterministically (i.e., coefficients of states that never appear as the final state of a transition). Such contamination is undesirable because it can produce artificial population oscillations. To avoid this, one may use a weighted normalization scheme. At each time step, the scheme normalizes the coefficients of the active states (i.e., states that participate in at least one transition as a final state) and the coefficients of the fixed states (i.e., states that do not participate in any transition as a final state) as follows:

$$\begin{aligned} \{\bar{C}_{\lambda_f}(t)\}_{\lambda_f \in \mathcal{M}_f} &\rightarrow \{\bar{C}_{\lambda_f}(t)\}_{\lambda_f \in \mathcal{M}_f} \\ \{\bar{C}_{\lambda_a}(t)\}_{\lambda_a \in \mathcal{M}_a} &\rightarrow \left\{ \sqrt{\frac{1 - p_f(t)}{p_a(t)}} \bar{C}_{\lambda_a}(t) \right\}_{\lambda_a \in \mathcal{M}_a} \end{aligned} \quad (3.1)$$

where $\mathcal{M}_f$ and $\mathcal{M}_a$ denote the manifolds of fixed and active states, respectively, and $p_f(t) = \sum_{\lambda_f \in \mathcal{M}_f} |\bar{C}_{\lambda_f}(t)|^2$ and $p_a(t) = \sum_{\lambda_a \in \mathcal{M}_a} |\bar{C}_{\lambda_a}(t)|^2$ are the populations of the fixed and active states at time t . When all states are involved in transitions as final states, the weighted normalization scheme is equivalent to the typical normalization scheme.

Chapter 4

Integration Schemes

The integration schemes used for the implementation of the SSE are the Euler-Maruyama scheme, the Heun scheme, and the Runge-Kutta (4th-order) scheme. The Euler-Maruyama scheme is the simplest one and is obtained by directly implementing Eq. (2.19). The Heun integration scheme is given by:

$$\bar{\mathbf{C}}_{n+1} = \bar{\mathbf{C}}_n + \frac{\Delta t}{2} \left(D(\bar{\mathbf{C}}_n, t) + D(\tilde{\mathbf{C}}, t + \Delta t) \right) + \frac{1}{2} \left(S(\bar{\mathbf{C}}_n, t) + S(\tilde{\mathbf{C}}, t + \Delta t) \right), \quad (4.1)$$

where

$$D(\bar{\mathbf{C}}_n, t) = \left(-i\mathbf{H}_n - \frac{1}{2}\mathbf{\Gamma} \right) \bar{\mathbf{C}}_n, \quad (4.2)$$

$$S(\bar{\mathbf{C}}_n, t) = -i\sqrt{\Delta t}\mathbf{R}_n\bar{\mathbf{C}}_n, \quad (4.3)$$

$$\tilde{\mathbf{C}} = \bar{\mathbf{C}}_n + \Delta t D(\bar{\mathbf{C}}_n, t) + S(\bar{\mathbf{C}}_n, t). \quad (4.4)$$

The Runge-Kutta (4th-order) scheme is given by:

$$\bar{\mathbf{C}}_{n+1} = \bar{\mathbf{C}}_{det} + S(\bar{\mathbf{C}}_{det}, t + \Delta t), \quad (4.5)$$

where

$$\bar{\mathbf{C}}_{det} = \bar{\mathbf{C}}_n + \frac{\Delta t}{6} (K_1 + 2K_2 + 2K_3 + K_4), \quad (4.6)$$

$$K_1 = D(\bar{\mathbf{C}}_n, t), \quad (4.7)$$

$$K_2 = D\left(\bar{\mathbf{C}}_n + \frac{\Delta t}{2}K_1, t + \frac{\Delta t}{2}\right), \quad (4.8)$$

$$K_3 = D\left(\bar{\mathbf{C}}_n + \frac{\Delta t}{2}K_2, t + \frac{\Delta t}{2}\right), \quad (4.9)$$

$$K_4 = D(\bar{\mathbf{C}}_n + \Delta t K_3, t + \Delta t). \quad (4.10)$$

Chapter 5

Implementation

5.1 Electric Field

The electric field used in all tests consists of a Gaussian pulse with a full width at half maximum (FWHM) of 9.4 as and an intensity of 1×10^4 W/cm². The time at which the electric field reaches its maximum is 1.0 atomic units (a.u.). The time step for the propagation is $\Delta t = 5 \times 10^{-3}$ a.u.

5.2 First Test: Two-State Model

For this first test, a two-state model with $\{|0\rangle, |1\rangle\}$ will be used. The only transition operator considered in this implementation is $\hat{S}_{1,0} = \sqrt{\Gamma_{0,1}} |0\rangle \langle 1|$, where $\Gamma_{0,1} = 2.0$. The averages over the stochastic noise are computed by averaging over 30 paths. The initial population of state $|0\rangle$ is 0.0 and the initial population of state $|1\rangle$ is 1.0.

For this model, the theoretical predictions for the populations of each state are computed using the following equation:

$$\begin{aligned} p_0(t) &= 1 - p_1(0)e^{-\Gamma_{0,1}t} \\ p_1(t) &= p_1(0)e^{-\Gamma_{0,1}t}, \end{aligned} \tag{5.1}$$

where $p_0(0) = 0.0$ and $p_1(0) = 1.0$.

The model results obtained with each integration scheme as well as the theoretical predictions are shown in Figures 1, 2 and 3 respectively.

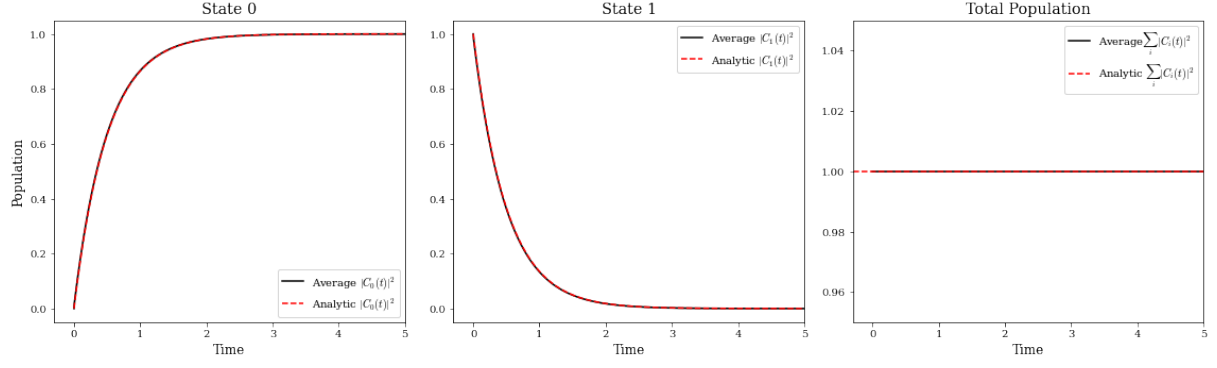


Figure 1: Results for the implementation of the stochastic Schrödinger equation (SSE) using the two-state model with a time step of $\Delta t = 5 \times 10^{-3}$ and the Euler-Maruyama scheme. Theoretical predictions are given by the dashed red lines.

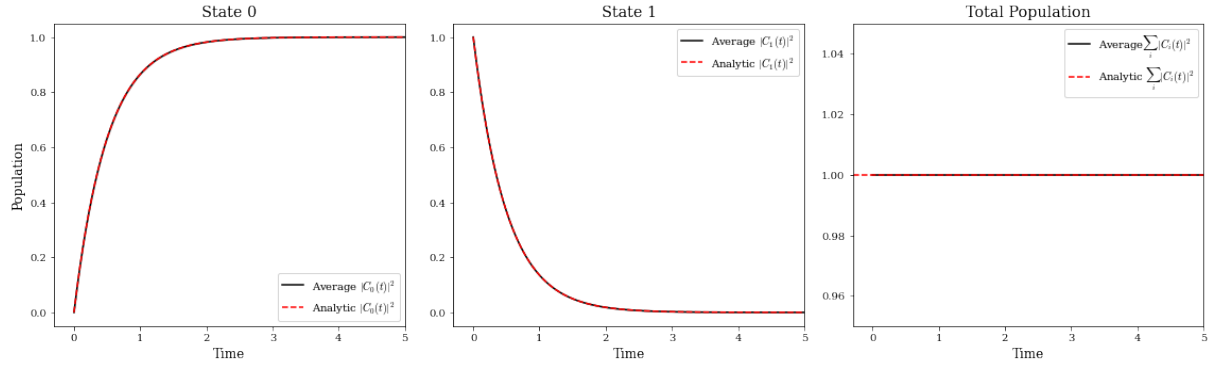


Figure 2: Results for the implementation of the stochastic Schrödinger equation (SSE) using the two-state model with a time step of $\Delta t = 5 \times 10^{-3}$ and the Heun scheme. Theoretical predictions are given by the dashed red lines.

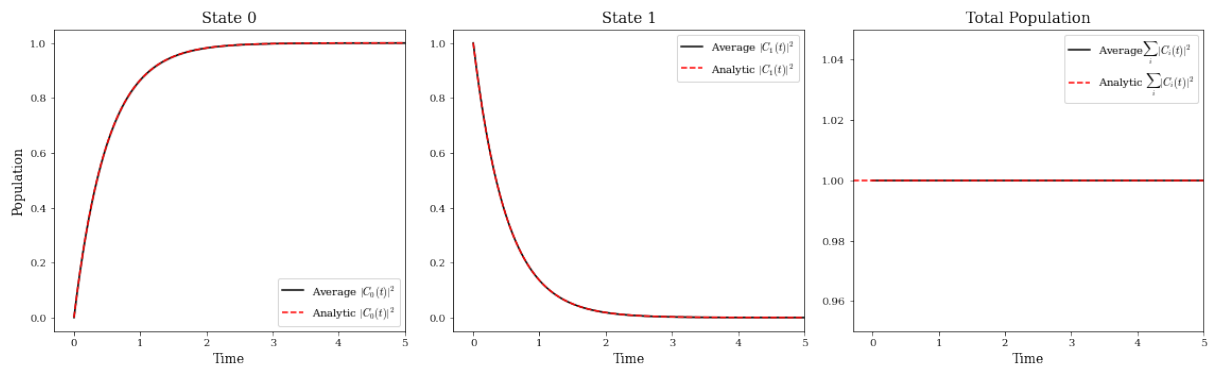


Figure 3: Results for the implementation of the stochastic Schrödinger equation (SSE) using the two-state model with a time step of $\Delta t = 5 \times 10^{-3}$ and the Runge-Kutta (4th-order) scheme. Theoretical predictions are given by the dashed red lines.

5.3 Second Test: Three-State Model (Two Transition Operators)

For the second test, a three-state model with $\{|0\rangle, |1\rangle, |2\rangle\}$ will be used. The only transition operators considered in this implementation are $\hat{S}_{1,0} = \sqrt{\Gamma_{0,1}} |0\rangle\langle 1|$ and $\hat{S}_{2,1} = \sqrt{\Gamma_{0,2}} |0\rangle\langle 2|$, where $\Gamma_{0,1} = 1.0$ and $\Gamma_{0,2} = 2.0$. The averages over the stochastic noise are computed by averaging over 30 paths. The initial population of state $|0\rangle$ is 0.0, the initial population of state $|1\rangle$ is 0.5 and the initial population of state $|2\rangle$ is 0.5.

For this model, the theoretical predictions for the populations of each state are computed using the following equation:

$$\begin{aligned} p_0(t) &= 1 - p_1(0)e^{-\Gamma_{0,1}t} - p_2(0)e^{-\Gamma_{0,2}t} \\ p_1(t) &= p_1(0)e^{-\Gamma_{0,1}t} \\ p_2(t) &= p_2(0)e^{-\Gamma_{0,2}t}, \end{aligned} \tag{5.2}$$

where $p_0(0) = 0.0$, $p_1(0) = 0.5$ and $p_2(0) = 0.5$.

The model results obtained with each integration scheme as well as the theoretical predictions are shown in Figures 4, 5 and 6 respectively.

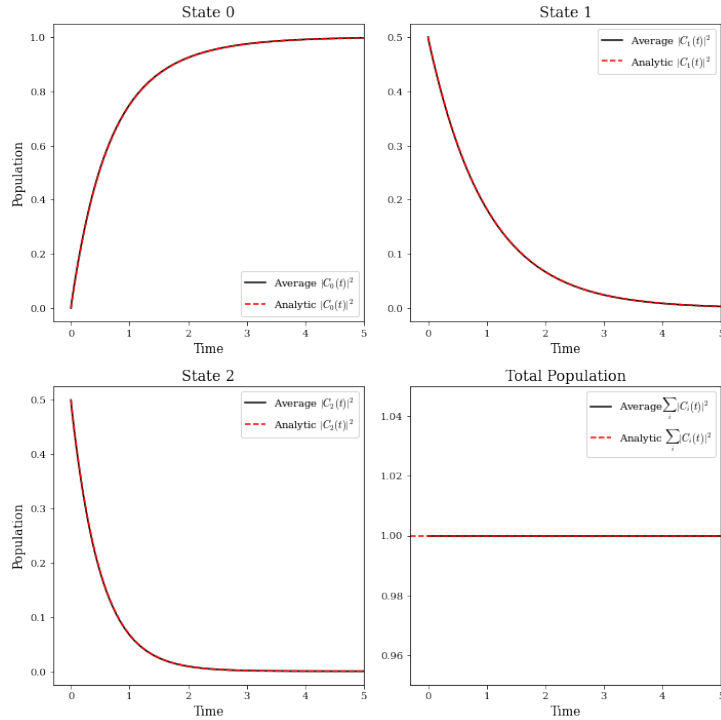


Figure 4: Results for the implementation of the stochastic Schrödinger equation (SSE) using a three-state model with two transition operators with a time step of $\Delta t = 5 \times 10^{-3}$ and the Euler-Maruyama scheme. Theoretical predictions are given by the dashed red lines.

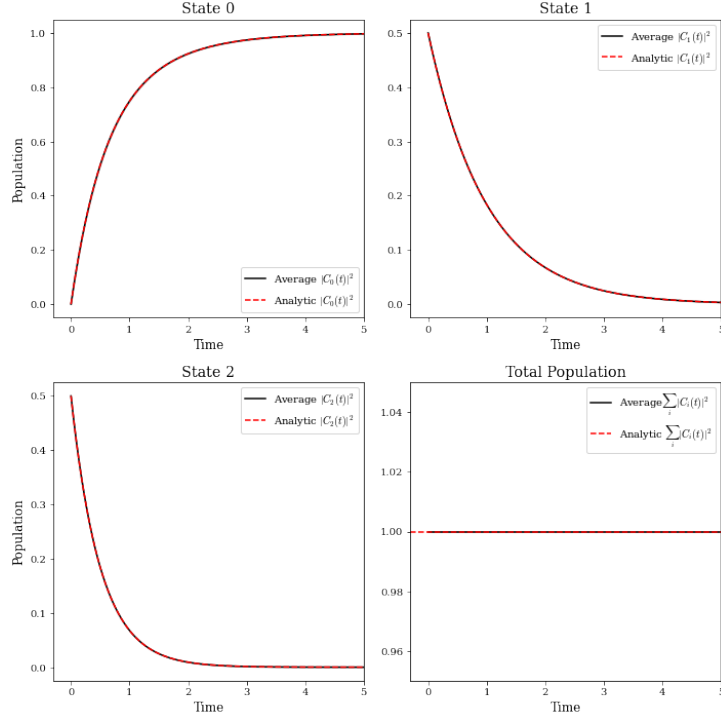


Figure 5: Results for the implementation of the stochastic Schrödinger equation (SSE) using a three-state model with two transition operators with a time step of $\Delta t = 5 \times 10^{-3}$ and the Heun scheme. Theoretical predictions are given by the dashed red lines.

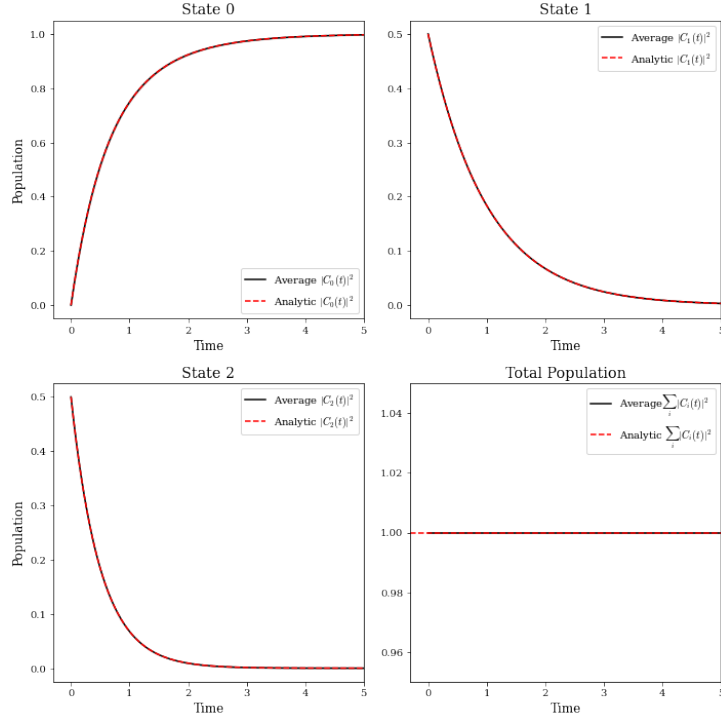


Figure 6: Results for the implementation of the stochastic Schrödinger equation (SSE) using a three-state model with two transition operators with a time step of $\Delta t = 5 \times 10^{-3}$ and the Runge-Kutta (4th-order) scheme. Theoretical predictions are given by the dashed red lines.

5.4 Third Test: Three-State Model (Single Transition Operator)

For the third test, a three-state model with $\{|0\rangle, |1\rangle, |2\rangle\}$ will be used. The only transition operator considered in this implementation is $\hat{S}_{2,1} = \sqrt{\Gamma_{1,2}} |1\rangle\langle 2|$. The averages over the stochastic noise are computed by averaging over 100 paths. The initial population of state $|0\rangle$ is 0.9, the initial population of state $|1\rangle$ is 0.0 and the initial population of state $|2\rangle$ is 0.1.

For this model, the theoretical predictions for the populations of each state are computed using the following equation:

$$\begin{aligned} p_0(t) &= 0.9 \\ p_1(t) &= 1 - p_0(0) - p_2(0)e^{-\Gamma_{1,2}t} \\ p_2(t) &= p_2(0)e^{-\Gamma_{1,2}t}, \end{aligned} \quad (5.3)$$

where $p_0(0) = 0.9$, $p_1(0) = 0$ and $p_2(0) = 0.1$.

The model results obtained with each integration scheme as well as the theoretical predictions are shown in Figures 7, 8 and 9 respectively.

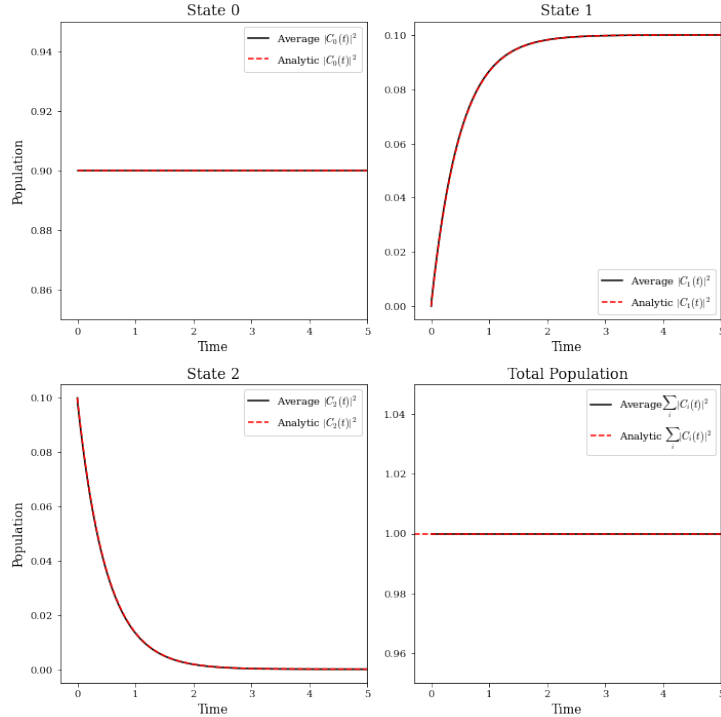


Figure 7: Results for the implementation of the stochastic Schrödinger equation (SSE) using a three-state model with a single transition operator with a time step of $\Delta t = 5 \times 10^{-3}$ and the Euler-Maruyama scheme. Theoretical predictions are given by the dashed red lines.

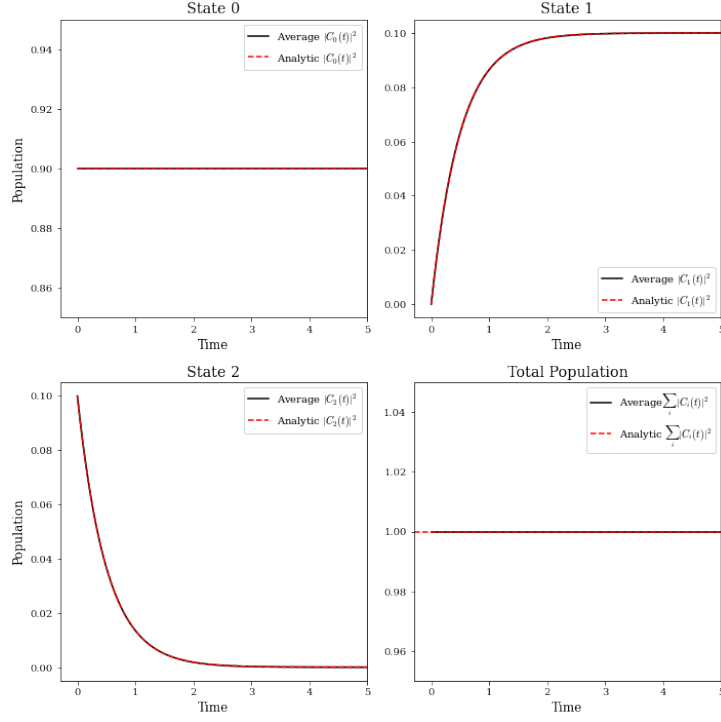


Figure 8: Results for the implementation of the stochastic Schrödinger equation (SSE) using a three-state model with a single transition operator with a time step of $\Delta t = 5 \times 10^{-3}$ and the Heun scheme. Theoretical predictions are given by the dashed red lines.

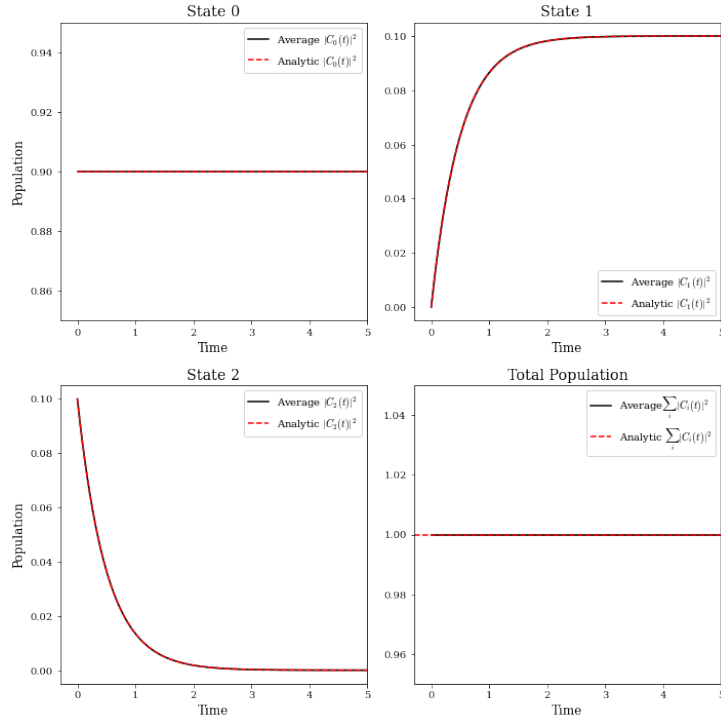


Figure 9: Results for the implementation of the stochastic Schrödinger equation (SSE) using a three-state model with a single transition operator with a time step of $\Delta t = 5 \times 10^{-3}$ and the Runge-Kutta (4th-order) scheme. Theoretical predictions are given by the dashed red lines.

5.5 Fourth Test: Vacuum LiCN

For the final test, a system of 11 states is used to simulate the propagation of a LiCN molecule in vacuum. The only transition operators considered in this implementation are $\hat{S}_{1,0} = \sqrt{\Gamma_{0,1}} |0\rangle \langle 1|$ and $\hat{S}_{2,0} = \sqrt{\Gamma_{0,2}} |0\rangle \langle 2|$, where $\Gamma_{0,1} = 0.001$ and $\Gamma_{0,2} = 0.004$. The averages over the stochastic noise are computed by averaging over 102 paths. The initial populations of the first six states are $1/6$, and the initial populations of the last five states are 0.0. This system was used as a test for the quantum jump method, implemented also in WaveT. The results comparing the Euler-Maruyama, Heun, and Runge-Kutta (4th-order) schemes with the quantum jump method are shown in Figure 10 for a time step of $\Delta t = 5 \times 10^{-2}$.

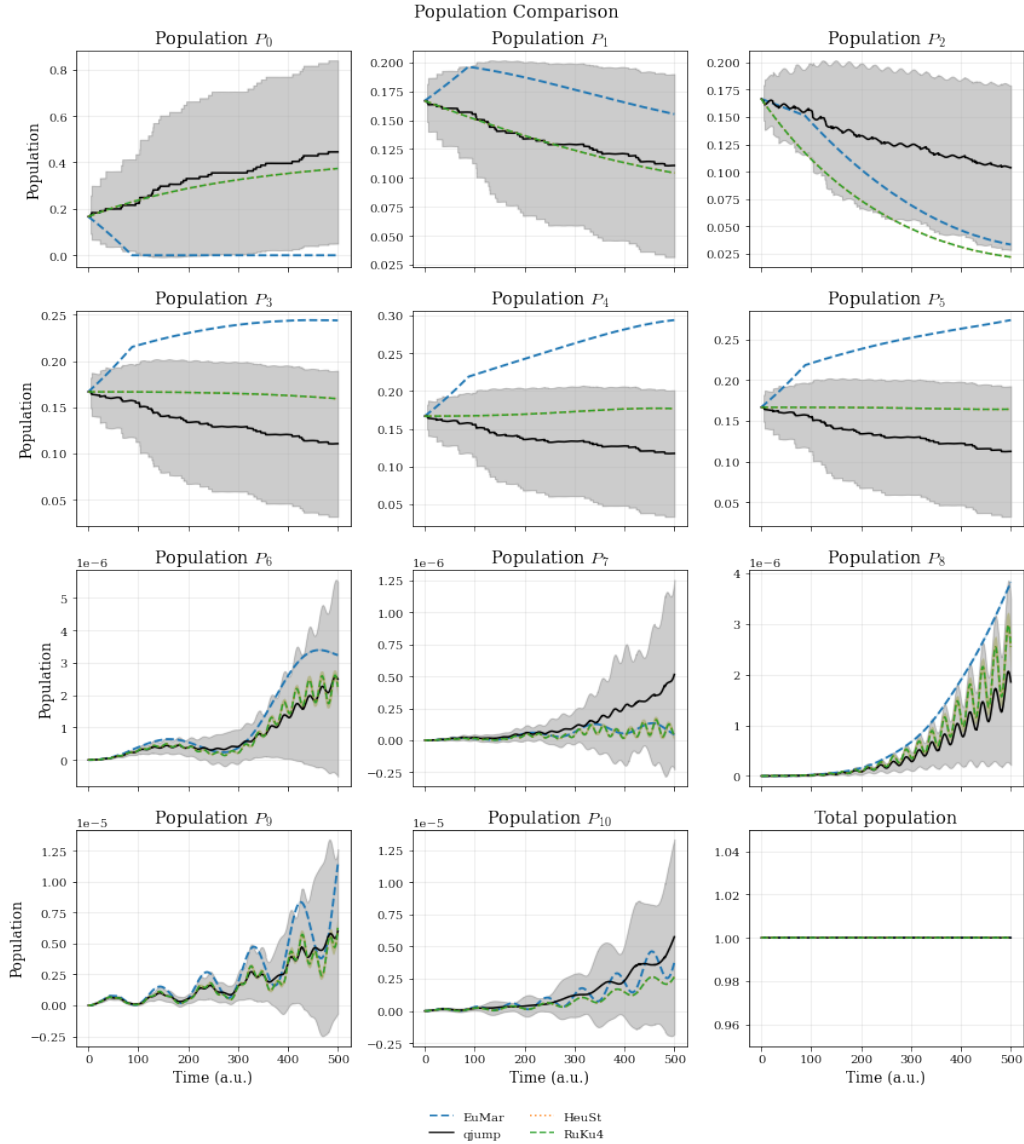


Figure 10: Results for the implementation of the stochastic Schrödinger equation (SSE) using a system of 11 states with a time step of $\Delta t = 5 \times 10^{-2}$ for all schemes considered in this section (Euler-Maruyama, Heun, Runge-Kutta (4th-order), and quantum jump method). The Euler-Maruyama, Heun, and Runge-Kutta (4th-order) schemes are compared with the quantum jump method.

The Euler-Maruyama scheme gave the worst results among the three schemes tested. For this reason, new results were obtained with the time step decreased to $\Delta t = 5 \times 10^{-3}$ only for this method. The results are shown in Figure 11.

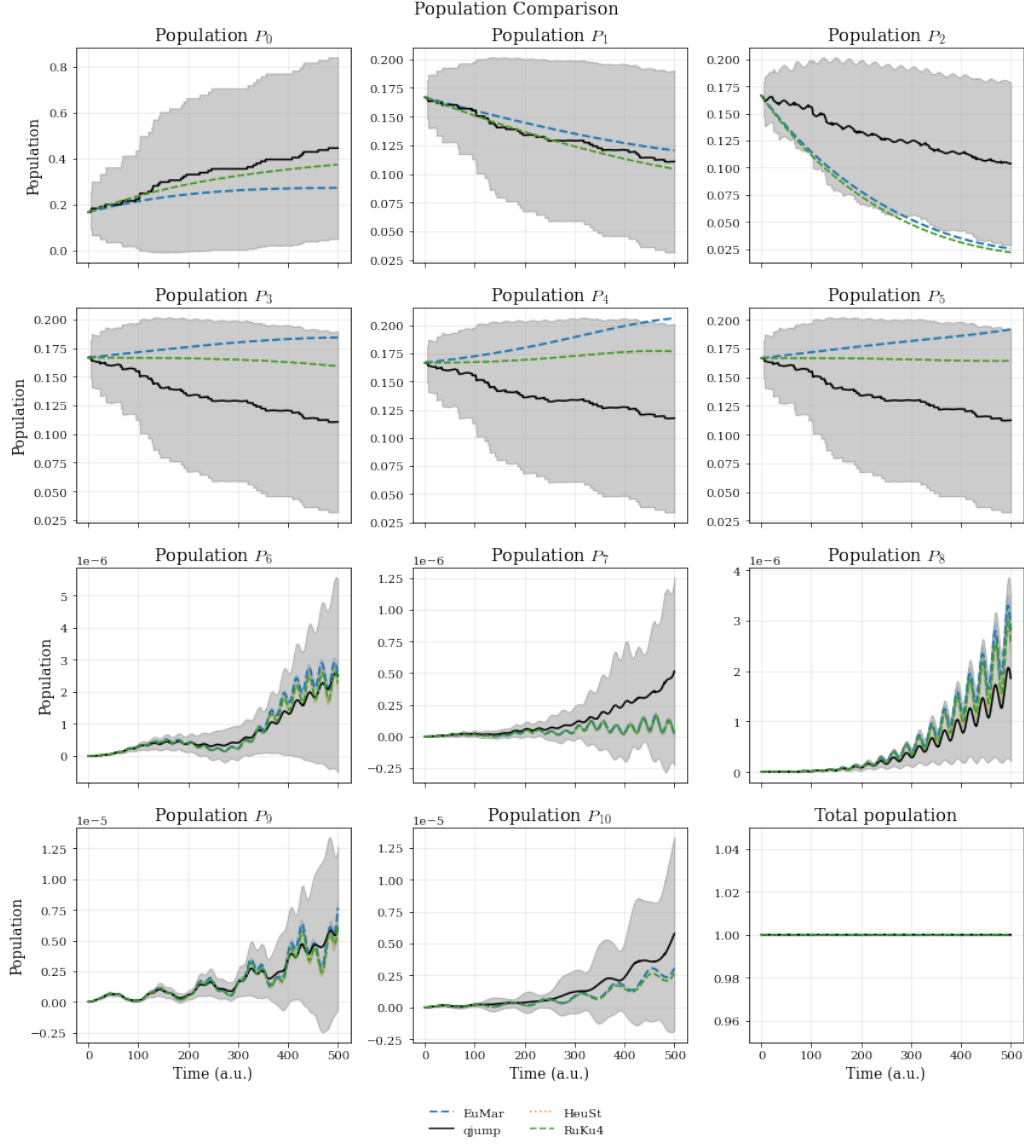


Figure 11: Results for the implementation of the stochastic Schrödinger equation (SSE) using a system of 11 states with a time step of $\Delta t = 5 \times 10^{-3}$ for the Euler-Maruyama scheme, and $\Delta t = 5 \times 10^{-2}$ for the other schemes.